Ana Silva

GIS & REMOTE SENSING · AGRICULTURE, WITH ATLASTICO

From pixels to *the full picture*.

Satellite analytics, large geospatial datasets and machine learning for deforestation, crops and carbon, mapped with a cartographer's eye.



ABOUT

From paper maps and cadastre to satellite time series and foundation models.

I grew up in Porto and now live in Sofia. I've drawn cartographic and cadastral maps, assessed environmental impacts, and spent the last several years building crop analytics, learning how crops behave across every continent and season.

Today I help organisations turn satellite imagery, open and commercial, into evidence they can act on.

HOW I CAN HELP

Environmental due diligence

Deforestation screening against the EUDR cutoff, protected areas, soil and climate exposure.

Crop & field monitoring

Crop type, phenology, anomalies and declaration checks, from one farm to a country.

Carbon & soil MRV

Practice evidence such as green cover between crops; persistent land-use change.

Cartography

Maps designed to be understood, for reports, publications and print.

PATH

- **Cartography & cadastre**
  where every boundary had to be right
- **Environmental assessment**
  impact studies and baselines
- **Agtech**
  several years of crop analytics at scale
- **Atlastico**
  crop, field and supply-chain analytics

WORKED WITH

Multinational food, beverage and retail companies, agricultural and forestry producers in Brazil, agricultural insurers, and the World Health Organization.

TOOLS

Python (xarray, dask, geopandas, scikit-learn, PyTorch) · QGIS · Google Earth Engine · STAC / COG

DATA

Sentinel-1/2/3/5P · Landsat · Maxar · Planet · parcel registers · Copernicus DEM

LANGUAGES

English · Portuguese · Bulgarian · Spanish

BASED IN

Sofia, Bulgaria

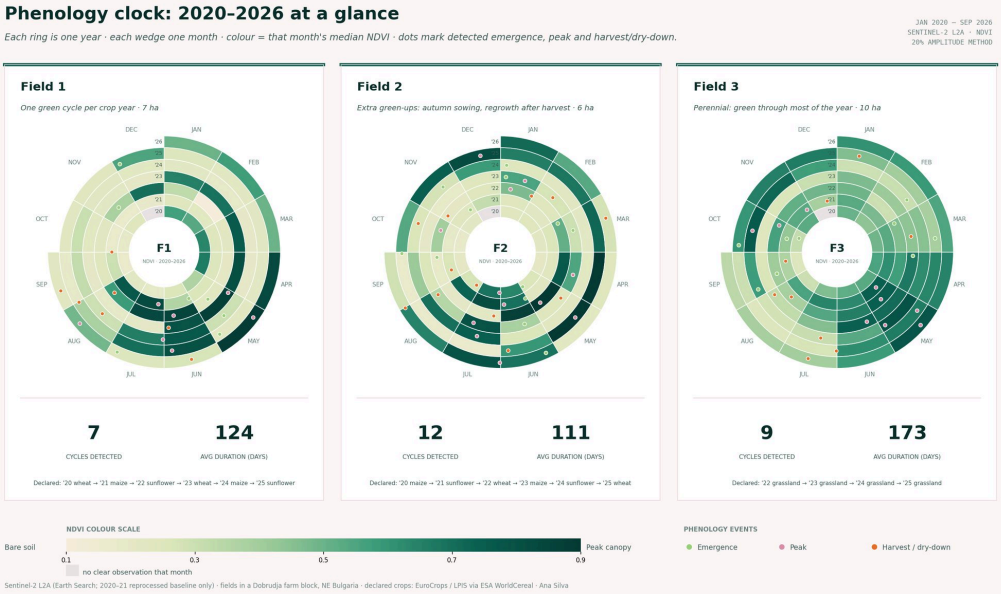
Crops, fields or supply chains?

Agriculture projects are taken on through Atlastico.

atlastico.com
linkedin.com/in/ana-luisa-silva
https://kirjawa.github.io/agriculture.html

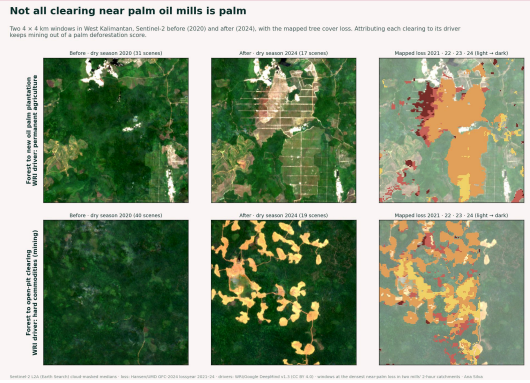
Selected work

Real Sentinel-1 and Sentinel-2 imagery · public parcel registers



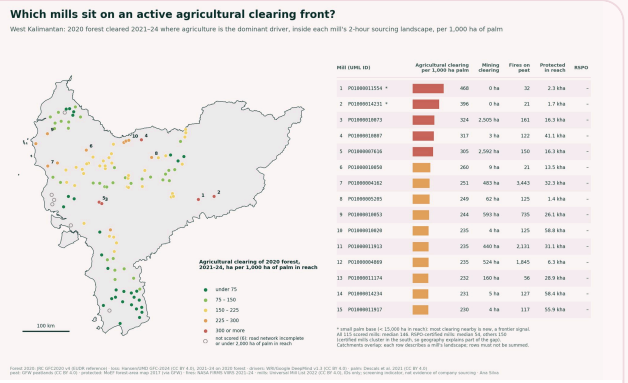
Phenology clock, seven years ring by ring

One ring per year on three neighbouring fields: rotations (wheat → maize → sunflower), emergence, peak and harvest, from 20,207 Sentinel-2 observations, 2020–2026.



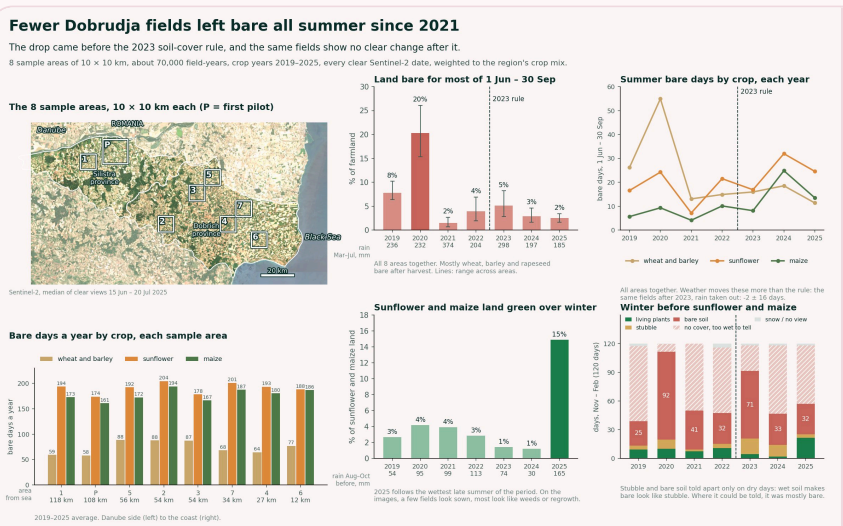
Palm oil mills: where does the fruit come from?

361 mills in West Kalimantan and Riau, their truck-reachable palm, and forest cleared since the EUDR cutoff, split by cause.



Mills on the clearing front

Agricultural clearing of 2020 forest per 1,000 ha of palm in reach. A screening signal, shown by mill ID.



Dobrudja: how long does the soil lie bare?

About 70,000 field-years in Bulgaria's grain belt, 2019–2025, every clear Sentinel-2 image. Fewer fields now sit bare all summer, but the drop came before the 2023 soil-cover rule, and winter is still the gap. Checked on the images.